



Basic Setup (Task I - TCP)

1. **Start the TCP lottery server:**

```
python3 lottery_server_tcp.py 127.0.0.1 5000
```

2. **Start clients (in separate terminals):**

```
python3 lottery_client_tcp_random.py 127.0.0.1 5000
```

UDP Version (Task I.b)

1. **Start UDP server:**

```
python3 lottery_server_udp.py 127.0.0.1 5001
```

2. **Client stays the same** (just change port to 5001)

Different Client Input Methods (Task IV)

- a) Random numbers (already shown above)

- b) **Standard input:**

```
python3 lottery_client_stdin.py 127.0.0.1 5000
# Then type: 1 5 10 15 20
# Then type: 50
```

- c) **JSON file:**

First create `picks.json` :

```
{  
  "numbers": [3, 7, 11, 15, 19],  
  "bet": 50  
}
```

Then run:

```
python3 lottery_client_json.py 127.0.0.1 5000 picks.json
```

d) Command line:

```
python3 lottery_client_cmdline.py 127.0.0.1 5000 2 8 12 16 20 75
```

History Server Versions (Task V)

V.a - Proxy Mode:

```
# Terminal 1: Start lottery server  
python3 lottery_server_tcp.py 127.0.0.1 5000  
  
# Terminal 2: Start history proxy (listens on UDP, forwards to lottery via TCP)  
python3 history_server_proxy.py 127.0.0.1 6000 127.0.0.1 5000 history_log.json  
  
# Terminal 3: Client connects to proxy (UDP)  
python3 lottery_client_proxy.py 127.0.0.1 6000
```

V.b - Separate History Server:

```
# Terminal 1: Start history server
python3 history_server_separate.py 127.0.0.1 7000 history_log.json

# Terminal 2: Start lottery server with history logging
python3 lottery_server_with_history.py 127.0.0.1 5000 127.0.0.1 7000

# Terminal 3: Regular client
python3 lottery_client_tcp_random.py 127.0.0.1 5000
```

V.c - Client Sends to History:

```
# Terminal 1: Start lottery server
python3 lottery_server_tcp.py 127.0.0.1 5000

# Terminal 2: Start history server
python3 history_server_separate.py 127.0.0.1 7000 history_log.json

# Terminal 3: Client that logs results
python3 lottery_client_with_history.py 127.0.0.1 5000 127.0.0.1 7000
```

Struct Version (Task II.b)

```
# Server with struct format
python3 lottery_server_struct.py 127.0.0.1 5000

# You'd need a matching struct client (not included in basic examples)
```

Checksum Version (Task II.c)

```
python3 lottery_server_checksum.py 127.0.0.1 5000  
# Needs matching checksum client
```

Quick Test:

The simplest way to test everything works:

```
# Terminal 1  
python3 lottery_server_tcp.py 127.0.0.1 5000  
  
# Terminal 2, 3, 4... (multiple clients)  
python3 lottery_client_tcp_random.py 127.0.0.1 5000
```

The server will handle all clients concurrently using `select()` !